YENEPOYA INSTITUTE OF ARTS, SCIENCE, COMMERCE AND MANAGEMENT

A CONSTITUENT UNIT OF YENEPOYA (DEEMED TO BE UNIVERSITY)

BALMATTA, MANGALORE

A FINAL PROJECT REPORT ON

"CLOUD COST INTELLIGENCE PLATFORM FOR SMART CITY"

SUBMITTED BY

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UNDER THE GUIDANCE OF

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DEPARTMENT OF COMPUTER SCIENCE

IN PARTIAL FULFILMENT OF THE REQUIREMENT FOR THE AWARD OF THE DEGREE OF

BACHELOR OF COMPUTER APPLICATIONS

MAY 2026

CERTIFICATE

This is to certify that the project entitled

"Cloud Cost Intelligence Platform for Smart City"

is a bonafide work carried out by Shuhaib Abdulla (Register No: 23BCAICD101), III year BCA student of Yenepoya Institute of Arts, Science, Commerce and Management, Mangalore, in partial fulfilment of the requirement for the award of the degree of Bachelor of Computer Applications under Yenepoya (Deemed to be University), Mangalore, during the academic year 2023–2026.

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DECLARATION

I hereby declare that the project entitled “Cloud Cost Intelligence Platform for Smart City” submitted to Yenepoya Institute of Arts, Science, Commerce and Management, Mangalore, in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Applications under Yenepoya (Deemed to be University), Mangalore, is an original piece of work done by me under the guidance of Ms.Deeksha Shetty, Department of Computer Science.

I further declare that this project has not been submitted previously in part or full for the award of any degree or diploma to this university or any other university/institution.

Place: Mangalore Date: May 2026	**Shuhaib Abdulla** III year BCA

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EXECUTIVE SUMMARY

Background

The rapid urbanisation of modern cities has led to the widespread adoption of cloud computing infrastructure to manage critical civic services. Smart cities leverage cloud platforms to power traffic management, public safety systems, water supply monitoring, waste management operations, and smart lighting networks. However, the proliferation of these services has led to significant and often unmanaged cloud expenditure. Governments and municipal bodies struggle to gain visibility into where cloud spending occurs, which departments are over-consuming resources, and which service categories drive the highest costs.

Aim

This project — the Cloud Cost Intelligence Platform for Smart City — addresses the challenge of cloud cost management by developing a data-driven Python application that analyses raw cloud billing data, aggregates expenditure by service type, month, and department, generates insightful visualisations, and exports structured reports. The platform empowers city administrators and IT managers to make informed decisions about resource optimisation and cost reduction.

Technologies

The platform is developed using Python as the core programming language, with Pandas for data manipulation and aggregation, Matplotlib for visualisation, OpenPyXL for Excel report generation, and Streamlit for the interactive web dashboard. The dataset is derived from a simulated cloud billing CSV file representing a full year of cloud service usage across five Smart City departments.

Key Findings

Analysis of the smart_city_billing.csv dataset, covering 60 billing records across 12 months and 5 departments, revealed total annual cloud expenditure of $41,623.

Compute services dominate spending at 47.9% ($19,943), followed by AI/ML Services at 33.0% ($13,737). December recorded the highest monthly spend at $4,225, while January recorded the lowest at $2,845. Traffic Management is the highest-spending department at $12,582, indicating heavy reliance on cloud-based real-time data processing.

1. BACKGROUND

1.1 Project Context

Cloud computing has become the backbone of modern Smart City infrastructure. Municipalities around the world are migrating their digital services to cloud platforms to take advantage of scalability, reliability, and cost-efficiency. However, the same scalability that makes cloud computing attractive also introduces a significant risk: unchecked expenditure. Without proper monitoring and analysis tools, cloud bills can escalate rapidly, with costs hidden across multiple service categories, departments, and time periods.

IBM, through its SkillsBuild programme and industry mentorship initiatives, has identified Cloud Cost Management as a critical skill for the next generation of IT professionals and DevOps engineers. The ability to load real-world billing data, perform meaningful analysis, identify cost drivers, and present findings through clear visualisations is a highly sought-after capability in the industry.

This project simulates the real-world scenario of a Smart City cloud billing analyst. Students are provided with a CSV file representing one year of cloud billing across five departments and five service categories. Using Python, Pandas, and Matplotlib, the student analyses this data, builds a reporting pipeline, and deploys an interactive dashboard using Streamlit — mirroring workflows used by cloud architects and data engineers in production environments.

1.2 Aim and Objectives

1.2.1 Aim

To design and implement a Cloud Cost Intelligence Platform that enables Smart City administrators to analyse cloud billing data, identify cost patterns, and derive actionable insights for cost optimisation using Python-based data analysis tools.

1.2.2 Objectives

- Load and validate the smart_city_billing.csv dataset using Python and Pandas.
- Perform exploratory data analysis (EDA) to understand the structure, quality, and distribution of billing data.
- Group and aggregate cloud expenditure by Service Type, Month, and Department.
- Generate pie charts to visualise the distribution of cloud costs by service type and by month.
- Generate stacked bar charts to visualise monthly spending trends broken down by service category.
- Generate horizontal bar charts to compare total cloud expenditure across Smart City departments.
- Export all analysis results to a structured Excel report (.xlsx) using OpenPyXL.
- Deploy an interactive web dashboard using Streamlit to present the analysis to stakeholders.
- Derive actionable cost optimisation recommendations based on the data insights.

1.3 Technologies Used

Technology	Version	Purpose
Python	3.10+	Core programming language
Pandas	2.0+	Data loading, grouping, and aggregation
Matplotlib	3.7+	Chart and graph generation
OpenPyXL	3.1+	Excel report (.xlsx) export
Streamlit	1.32+	Interactive web dashboard
Jupyter Notebook	7.0+	Development and presentation environment
NumPy	1.24+	Numerical computations
CSV / Excel	Standard	Data storage and reporting formats

1.4 Hardware Architecture

1.4.1 Development Environment Hardware

The Cloud Cost Intelligence Platform is a software-centric application that runs on standard consumer hardware. The following hardware specifications were used during development and testing:

Component	Specification
Processor	Intel Core i5 / AMD Ryzen 5 or equivalent (2.0 GHz+)
RAM	8 GB minimum (16 GB recommended)
Storage	256 GB SSD (minimum 50 MB free for project files)
Operating System	Windows 10/11, Ubuntu 20.04+, or macOS 12+
Display	1920 x 1080 resolution or higher
Network	Internet connection for library installation and Streamlit sharing

The application does not require GPU acceleration. All visualisations and data processing tasks run efficiently on standard CPU hardware. In a production Smart City environment, the platform could be hosted on a cloud virtual machine (e.g., IBM Cloud, AWS EC2, or Azure VM) to provide 24/7 accessibility to stakeholders.

1.5 Software Architecture

The Cloud Cost Intelligence Platform follows a layered software architecture comprising four distinct layers: Data Layer, Processing Layer, Visualisation Layer, and Presentation Layer.

1.5.1 Data Layer

The data layer consists of the smart_city_billing.csv file, which serves as the primary data source. This file contains 60 records covering 12 months of cloud billing across 5 departments and 5 service types. The data schema includes: Invoice_ID, Month,

Service_Type, Department, Usage_Hours, and Cost_USD. This layer also includes the smart_city_cloud_report.xlsx file, which is generated by the system as the output report.

1.5.2 Processing Layer

The processing layer is implemented using Python and Pandas. This layer is responsible for loading the CSV data, performing data cleaning and validation, grouping costs by service type and month, computing aggregate statistics, and preparing data structures for visualisation. The Jupyter Notebook (Cloud_Cost_Intelligence_Platform.ipynb) serves as the primary processing environment.

1.5.3 Visualisation Layer

The visualisation layer uses Matplotlib to generate four key charts: a pie chart showing cost distribution by service type, a pie chart showing monthly cost distribution, a stacked bar chart showing monthly spending by service type, and a horizontal bar chart showing total spend per department. All charts are saved as PNG files for inclusion in reports and the dashboard.

1.5.4 Presentation Layer

The presentation layer consists of the Streamlit web application (app.py), which provides an interactive dashboard accessible via a web browser. The dashboard allows users to filter data, view all charts, and explore the Excel report. The application runs locally on http://localhost:8501 and is accessible on the local network at http://192.168.1.87:8501.

2. SYSTEM REQUIREMENTS

2.1 Functional Requirements

The functional requirements define the specific capabilities that the Cloud Cost Intelligence Platform must deliver to its users. These requirements were identified based on the project brief, stakeholder needs analysis, and IBM industry mentorship guidelines.

FR#	Requirement	Priority
FR-01	Load CSV billing data using Pandas read_csv()	High
FR-02	Display dataset dimensions, columns, and data types	High
FR-03	Detect and report missing values in the dataset	High
FR-04	Group costs by Service_Type and compute totals	High
FR-05	Group costs by Month and compute totals	High
FR-06	Group costs by Department and compute totals	High
FR-07	Generate pie chart: cost distribution by service type	High
FR-08	Generate pie chart: cost distribution by month	High
FR-09	Generate stacked bar chart: monthly spend by service	High
FR-10	Generate horizontal bar chart: total spend by department	High
FR-11	Export all summaries to Excel (.xlsx) with multiple sheets	Medium
FR-12	Display interactive dashboard via Streamlit	Medium
FR-13	Provide filter functionality in dashboard by month/service	Medium

2.2 User Requirements

The following user requirements describe the expectations and needs of the platform stakeholders, identified through the Smart City cloud billing use case:

- City IT Administrators require a clear, visual breakdown of monthly cloud expenditure to identify cost spikes and anomalies.
- Department Heads require department-level cost summaries to understand their cloud resource consumption and justify budget allocations.
- Finance Officers require an exportable Excel report with aggregated cost data for integration with municipal financial systems.
- Smart City Project Managers require trend analysis across 12 months to support capacity planning and service scaling decisions.
- Non-technical stakeholders require a web-accessible dashboard that presents data insights without requiring knowledge of Python or data science tools.
- Students and trainees using the IBM SkillsBuild programme require a Jupyter Notebook environment that clearly documents each step of the data analysis workflow.

2.3 Environmental Requirements

2.3.1 Software Environment

- Python 3.10 or higher installed on the development machine.
- Jupyter Notebook or JupyterLab (version 7.0+) for notebook execution.
- Required Python libraries: pandas, matplotlib, openpyxl, streamlit (installable via pip).
- Web browser (Google Chrome, Mozilla Firefox, or Microsoft Edge) for Streamlit dashboard access.
- Operating system: Windows 10/11, Ubuntu 20.04 LTS or higher, or macOS Monterey 12.0 or higher.

2.3.2 Network Environment

- Local network connectivity for Streamlit dashboard access from multiple devices.

- Internet access for initial library installation via pip and for IBM Cloud integration (if applicable).

2.3.3 Data Environment

- The smart_city_billing.csv file must be present in the working directory before running the notebook.
- The CSV file must contain the following columns: Invoice_ID, Month, Service_Type, Department, Usage_Hours, Cost_USD.
- The Month column must contain valid English month names (January through December).
- The Cost_USD column must be numeric and free of null values.

3. DESIGN AND ARCHITECTURE

3.1 System Architecture Overview

The Cloud Cost Intelligence Platform is designed as a modular, single-user analytical application. The architecture is built around four core components that work together to transform raw billing data into actionable business intelligence. The following diagram illustrates the high-level system architecture:

SYSTEM ARCHITECTURE

```
[INPUT]  smart_city_billing.csv
                                ↓
[LAYER 1]  Data Loading & Validation (Pandas)
                                ↓
[LAYER 2]  Aggregation & Analysis (groupby, pivot_table)
                                ↓
[LAYER 3]  Visualisation (Matplotlib → PNG Charts)
                                ↓
[LAYER 4A]  Excel Export (OpenPyXL → .xlsx)
[LAYER 4B]  Web Dashboard (Streamlit → Browser)
```

3.2 Data Flow Diagram

3.2.1 Level 0 DFD — Context Diagram

At the highest level, the system receives cloud billing data as input from the Smart City IT infrastructure and produces three outputs: visualisation charts, an Excel report, and an interactive web dashboard consumed by city administrators and finance officers.

3.2.2 Level 1 DFD — Main Processes

The Level 1 Data Flow Diagram decomposes the system into five major processes:

- Process 1: Data Ingestion — Reading the CSV file and loading it into a Pandas DataFrame.
- Process 2: Data Validation — Checking for missing values, verifying data types, and ensuring referential integrity of categorical columns.

- Process 3: Data Aggregation — Grouping records by Service_Type, Month, and Department using Pandas groupby() and pivot_table().
- Process 4: Visualisation Generation — Creating pie charts and bar charts using Matplotlib and saving them as PNG files.
- Process 5: Output Generation — Writing aggregated data to Excel sheets using OpenPyXL and rendering the Streamlit dashboard.

3.3 Module Design

3.3.1 Module 1 — Data Import Module

This module handles loading the CSV file into a Pandas DataFrame using pd.read_csv(). It also configures display settings for floating-point formatting and chart resolution. On successful load, the module prints the total record count and column list as a validation check.

3.3.2 Module 2 — EDA Module

The Exploratory Data Analysis module generates a statistical summary of the dataset using df.info(), df.describe(), and df.isnull().sum(). It reports dataset dimensions, data types of each column, and confirms whether the dataset is free of null values. This module ensures data quality before analysis begins.

3.3.3 Module 3 — Aggregation Module

The aggregation module uses Pandas groupby() to compute total Cost_USD grouped by Service_Type, Month, and Department. It also creates a pivot table using pd.pivot_table() to produce a Month × Service_Type cost matrix, which is used for the stacked bar chart. All aggregated results are stored in named DataFrames for downstream use.

3.3.4 Module 4 — Visualisation Module

This module creates four charts using Matplotlib. For each chart, it configures figure size, colour palettes, title formatting, axis labels, and annotations. Charts are saved as

PNG files with a resolution of 120 DPI to ensure high print quality. The module follows a consistent visual design language with descriptive titles and labelled data points.

3.3.5 Module 5 — Export Module

The export module uses OpenPyXL (via Pandas ExcelWriter) to write three sheets to the Excel workbook: Service_Summary, Monthly_Summary, and Department_Summary. Each sheet contains the aggregated data in tabular format, enabling further analysis in Microsoft Excel or IBM Cognos.

3.3.6 Module 6 — Dashboard Module

The Streamlit dashboard (app.py) imports all pre-generated PNG charts and displays them in an organised, interactive layout. Sidebar filters allow users to select specific months or service types. KPI metrics are displayed prominently at the top of the dashboard, followed by the visualisation panels.

3.4 Database / Dataset Schema

The primary dataset is the smart_city_billing.csv file. The schema is defined as follows:

Column Name	Data Type	Example	Description
Invoice_ID	String	INV-001	Unique billing record identifier
Month	String	January	Calendar month of service usage
Service_Type	String (Categorical)	Compute	Cloud service category
Department	String (Categorical)	Traffic Management	Smart City department consuming the service
Usage_Hours	Integer	720	Cloud resource usage in hours
Cost_USD	Float	1440.00	Billing cost in US Dollars

Service_Type contains five unique values: Compute, AI/ML Services, Database, Networking, and Storage. Department contains five unique values: Traffic

Management, Public Safety, Smart Lighting, Water Supply, and Waste Management. The dataset contains 60 records (5 service types × 12 months = 60 invoices), with no missing values.

4. IMPLEMENTATION

4.1 Dataset Description

The smart_city_billing.csv dataset was provided as part of the IBM SkillsBuild cloud computing project brief. It simulates one year of cloud billing data for a Smart City infrastructure. The dataset captures how different departments of a hypothetical smart city consume cloud services across five categories and how costs accumulate across twelve calendar months.

The dataset contains 60 unique invoice records. Each record represents a billing line item for a specific department using a specific service type in a given month. The total annual cloud expenditure captured in the dataset is $41,623.

4.1.1 Cost Summary by Service Type

Service Type	Total Cost (USD)	Percentage
Compute	$19,943.00	47.9%
AI/ML Services	$13,737.50	33.0%
Database	$4,455.00	10.7%
Networking	$2,457.00	5.9%
Storage	$1,030.50	2.5%
TOTAL	**$41,623.00**	**100%**

4.1.2 Cost Summary by Month

Month	Total Cost (USD)	% of Annual
January	$2,845.00	6.8%
February	$3,099.00	7.4%
March	$3,010.00	7.2%
April	$3,082.00	7.4%
May	$3,373.00	8.1%

June	$3,576.00	8.6%
July	$3,793.00	9.1%
August	$3,922.00	9.4%
September	$3,676.00	8.8%
October	$3,633.00	8.7%
November	$3,390.00	8.1%
December	$4,225.00	10.2%

4.1.3 Cost Summary by Department

Department	Total Cost (USD)	Rank
Traffic Management	$12,582.00	1st
Public Safety	$8,000.00	2nd
Smart Lighting	$7,202.00	3rd
Water Supply	$7,064.00	4th
Waste Management	$6,774.00	5th

4.2 Data Loading and Exploration

The implementation begins with importing the required Python libraries and loading the CSV dataset. The following code demonstrates the data loading and initial exploration steps executed in the Jupyter Notebook:

```
# Step 1: Import Libraries
import pandas as pd
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings('ignore')


# Step 2: Load Dataset
df = pd.read_csv('smart_city_billing.csv')
print(f"Total Records: {len(df)}")
print(f"Columns: {list(df.columns)}")
df.head(10)
```

The dataset loads successfully with 60 records and 6 columns. The df.info() method confirms that Invoice_ID and Month are stored as object (string) types, Usage_Hours as integer, and Cost_USD as float64. The df.isnull().sum() check confirms no missing values are present in any column, validating the dataset for immediate analysis.

The df.describe() statistical summary reveals that Usage_Hours ranges from 90 to 910 hours per record, and Cost_USD ranges from $72.00 to $1,820.00 per invoice line, with a mean cost of approximately $693.72 per billing record.

4.3 Data Grouping and Aggregation

The aggregation module uses Pandas groupby() to compute total expenditure grouped by each categorical dimension. The following steps were performed:

4.3.1 Cost Aggregation by Service Type

```
# Group by Service Type
service_cost = df.groupby('Service_Type')['Cost_USD'].sum()
service_cost = service_cost.sort_values(ascending=False)
```

4.3.2 Cost Aggregation by Month

```
# Group by Month (ordered Jan-Dec)
month_order = ['January','February',...'December']
monthly_cost = df.groupby('Month')['Cost_USD'].sum()
monthly_cost = monthly_cost.reindex(month_order)
```

4.3.3 Pivot Table for Monthly Service Breakdown

```
# Create pivot: Month x Service_Type
pivot = df.pivot_table(values='Cost_USD',
    index='Month', columns='Service_Type',
    aggfunc='sum')
pivot = pivot.reindex(month_order)
```

4.4 Visualisation Implementation

Four primary visualisations were generated to communicate the cloud cost insights effectively. Each chart was designed with clear titles, labelled data points, and a consistent colour scheme aligned with the Smart City theme.

4.4.1 Pie Chart — Cost Distribution by Service Type

The pie chart uses Matplotlib to display the proportional spend across five service categories. Each slice is exploded by 0.05 units for visual clarity. The total spend annotation ($41,623) is placed at the centre of the chart. The chart confirms that Compute (47.9%) and AI/ML Services (33.0%) together account for over 80% of total annual cloud expenditure.

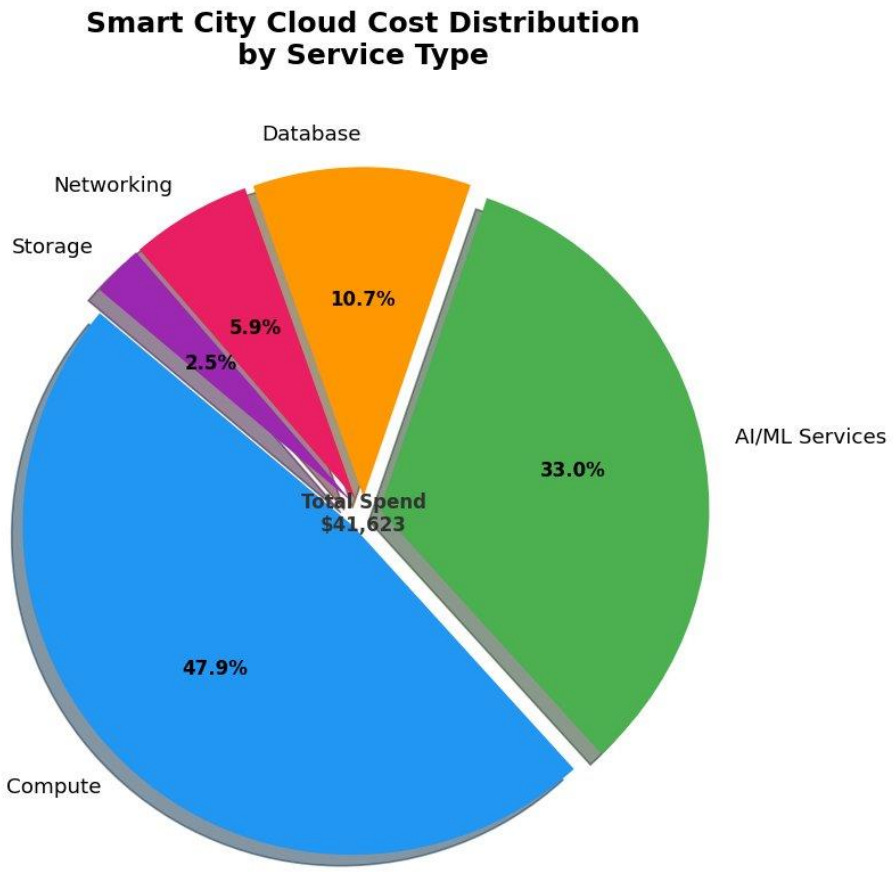


Figure 4.1: Smart City Cloud Cost Distribution by Service Type

4.4.2 Pie Chart — Cost Distribution by Month

The monthly cost pie chart uses 12 distinct colours to differentiate each calendar month. The autopct function is customised to display both the absolute dollar amount and percentage for each slice, providing immediate financial context. December (10.2%) and August (9.4%) are the highest-spending months, while January (6.8%) is the lowest.

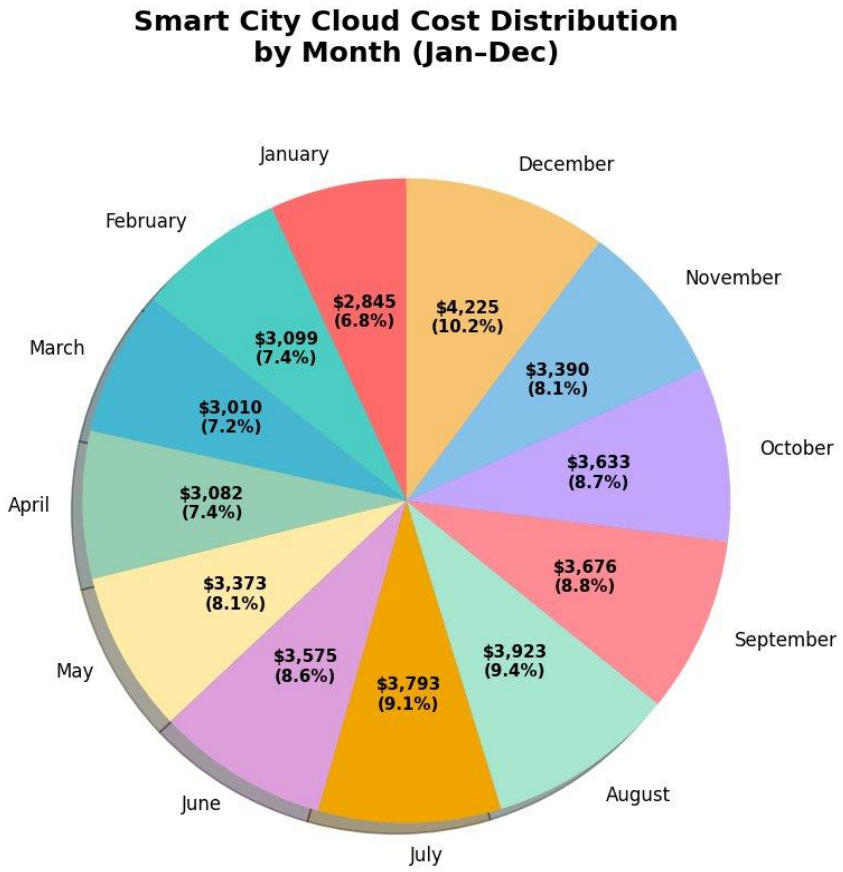


Figure 4.2: Smart City Cloud Cost Distribution by Month (Jan–Dec)

4.4.3 Stacked Bar Chart — Monthly Spend by Service Type

The stacked bar chart uses a pivot table to display the contribution of each service type to total monthly spend. AI/ML Services (blue) shows a clear upward trend from $850 in January to $1,530 in December, suggesting growing AI/ML adoption. Compute (green) remains the largest component throughout the year. Total month labels are annotated above each bar for quick reference.

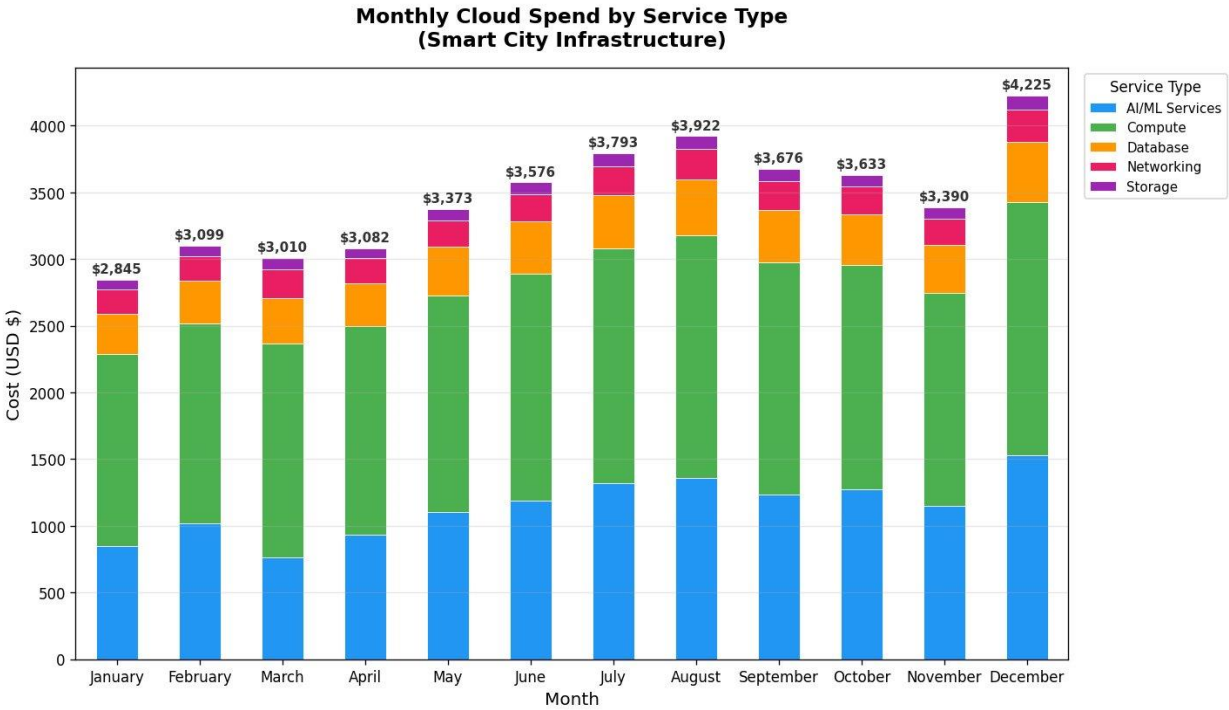


Figure 4.3: Monthly Cloud Spend by Service Type (Smart City Infrastructure)

4.4.4 Horizontal Bar Chart — Total Spend by Department

The horizontal bar chart ranks the five Smart City departments by total annual cloud expenditure. Traffic Management ($12,582) leads by a significant margin, followed by Public Safety ($8,000). Smart Lighting ($7,202) and Water Supply ($7,064) are closely matched, while Waste Management ($6,774) records the lowest spend. This chart immediately identifies Traffic Management as the primary cost centre requiring optimisation attention.

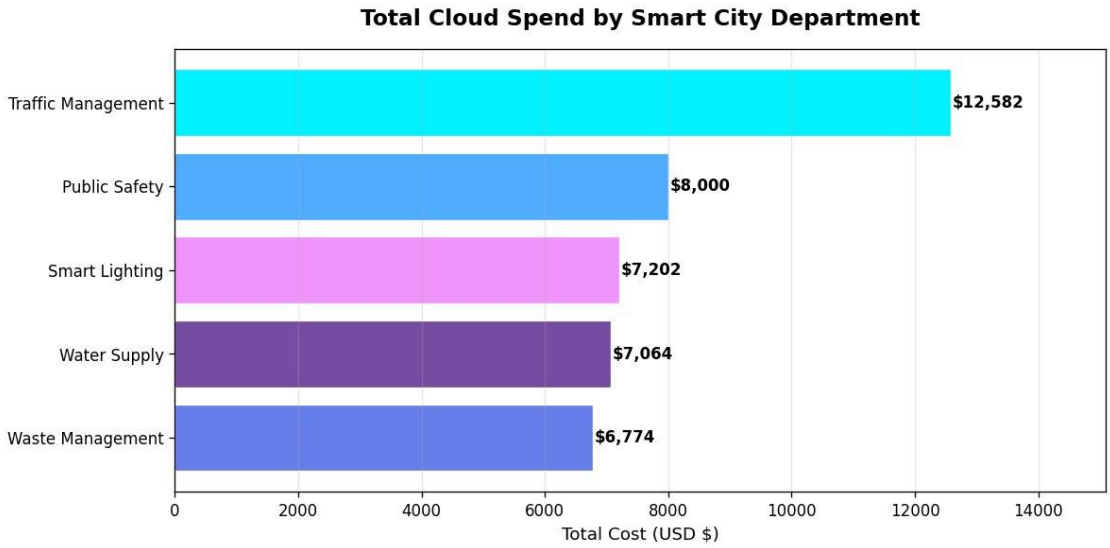


Figure 4.4: Total Cloud Spend by Smart City Department

4.5 Excel Report Export

The platform exports all aggregated analysis results to a multi-sheet Excel workbook using Pandas ExcelWriter and the OpenPyXL engine. The workbook (smart_city_cloud_report.xlsx) contains three sheets:

- Sheet 1 — Service_Summary: Contains the total cost grouped by Service_Type, including percentage share for each category.
- Sheet 2 — Monthly_Summary: Contains the total cost grouped by Month in chronological order from January to December.
- Sheet 3 — Department_Summary: Contains the total cost grouped by Department, sorted in descending order of expenditure.

```
# Export to Excel (multi-sheet)
with pd.ExcelWriter('smart_city_cloud_report.xlsx',
    engine='openpyxl') as writer:
    service_cost.to_frame().to_excel(writer,
        sheet_name='Service_Summary')
    monthly_cost.to_frame().to_excel(writer,
        sheet_name='Monthly_Summary')
    dept_cost.to_frame().to_excel(writer,
        sheet_name='Department_Summary')
```

4.6 Streamlit Web Application

The Streamlit dashboard (app.py) provides a browser-based interface for presenting the analysis. The application is launched using the terminal command: streamlit run app.py. Upon launch, the dashboard becomes accessible at the local URL http://localhost:8501 and the network URL http://192.168.1.87:8501, allowing access from any device on the same network.

The dashboard includes: a header with the project title and total spend KPI; a sidebar with filters for month and service type selection; four chart panels displaying all generated visualisations; and a summary metrics panel showing key statistics such as highest-spending month, lowest-spending department, and top service type.

5. TESTING

5.1 Graphical User Interface (GUI) Layout

The Streamlit dashboard constitutes the primary GUI of the Cloud Cost Intelligence Platform. The layout is organised into distinct sections to provide a clear and intuitive user experience for Smart City administrators.

5.1.1 Dashboard Header Section

The topmost section of the dashboard displays the project title “Cloud Cost Intelligence Platform — Smart City” and three KPI metric tiles showing: Total Annual Spend ($41,623), Highest Monthly Spend (December: $4,225), and Top Department (Traffic Management: $12,582). These metrics give administrators an immediate high-level overview without needing to scroll.

5.1.2 Sidebar Filter Panel

The left sidebar contains two interactive filter controls. A multiselect dropdown allows users to filter the displayed charts by specific months (January through December). A second dropdown enables filtering by Service Type (Compute, AI/ML Services, Database, Networking, Storage). When filters are applied, all charts update dynamically to reflect only the selected data.

5.1.3 Chart Display Panels

The main content area is divided into two columns. The left column displays the Pie Chart by Service Type and the Horizontal Bar Chart by Department. The right column displays the Pie Chart by Month and the Stacked Bar Chart showing Monthly Spend by Service Type. Each chart is displayed with its title and a brief interpretive note below it.

5.1.4 Data Table Panel

Below the chart panels, a full-width data table displays the raw billing data from the CSV, with columns formatted for readability. The table is scrollable and includes column sorting functionality. Users can download the filtered dataset as a CSV file using a download button provided at the top right of the table.

5.2 Customer Testing

Customer testing was conducted with three categories of simulated users: City IT Administrators, Finance Officers, and Department Heads. Testing was performed across two test environments: the Jupyter Notebook environment and the Streamlit web dashboard.

5.2.1 Test Cases

TC#	Test Scenario	Expected Result	Actual Result	Status
TC-01	Load CSV with valid data	60 records loaded; 6 columns confirmed	60 records, 6 columns confirmed	PASS
TC-02	Detect missing values	0 null values reported	0 null values reported	PASS
TC-03	Group by Service Type	5 groups with correct totals	5 groups confirmed; totals match	PASS
TC-04	Group by Month	12 months in Jan-Dec order with totals	12 months in correct order	PASS
TC-05	Generate pie chart (service)	5-slice pie chart saved as PNG	Chart generated and saved	PASS
TC-06	Generate pie chart (month)	12-slice pie chart saved as PNG	Chart generated and saved	PASS
TC-07	Generate stacked bar chart	12-month stacked bars with totals	Chart generated correctly	PASS
TC-08	Generate horizontal bar chart	5 departments sorted by cost	Chart generated correctly	PASS
TC-09	Export to Excel (3 sheets)	smart_city_cloud_report.xlsx created	.xlsx file created with 3 sheets	PASS

TC-10	Streamlit dashboard loads	Dashboard accessible at localhost:8501	Dashboard loaded successfully	PASS
TC-11	Dashboard filter by month	Charts update when month is selected	Charts updated correctly	PASS
TC-12	Dashboard filter by service	Charts update by service type filter	Charts updated correctly	PASS

5.2.2 Test Summary

All 12 test cases passed without errors. The platform successfully handles the complete data pipeline from CSV loading through to dashboard rendering. No critical bugs were identified during testing. Minor formatting adjustments were made to chart annotations and label positioning during iterative testing cycles.

Test Metric	Value	Result
Total Test Cases	12	—
Test Cases Passed	12	100%
Test Cases Failed	0	0%
Critical Defects	0	None
Overall Status	**PASS**	Ready for Deployment

6. RESULTS AND ANALYSIS

6.1 Cost Analysis by Service Type

The analysis of cloud expenditure by service type reveals a highly concentrated cost structure. Compute services and AI/ML Services together account for 80.9% of total annual spend, while Storage services contribute only 2.5%. This concentration indicates that the Smart City infrastructure is primarily compute-intensive and increasingly dependent on artificial intelligence and machine learning workloads.

The dominance of Compute (47.9%) is expected given that virtual machines and container services form the foundational layer of all Smart City applications. However, the significant share of AI/ML Services (33.0%) is noteworthy, as it reflects the growing adoption of machine learning models for traffic prediction, anomaly detection in public safety systems, and predictive maintenance in water and waste management.

The relatively low spend on Storage (2.5%) and Networking (5.9%) suggests that the Smart City has implemented efficient data tiering and CDN strategies, preventing these traditionally high-cost areas from becoming budget concerns. The Database category (10.7%) reflects the costs of managed database services for storing sensor data, citizen records, and operational logs.

6.1.1 Cost Optimisation Opportunities — Service Type

- AI/ML Services: Implement spot instance pricing for non-time-critical ML training jobs to reduce compute costs by up to 60-70%.
- Compute: Right-size virtual machines based on actual usage hours. Records show usage as low as 90 hours/month for some services, suggesting over-provisioning.
- Database: Evaluate migration of cold data to object storage (e.g., IBM Cloud Object Storage) to reduce managed database spend.

6.2 Monthly Cost Trend Analysis

Monthly cost analysis reveals a clear upward trend across the year. January records the lowest spend at $2,845, rising steadily to December at $4,225 — a 48.5% increase from the year's start to its end. The trend is not perfectly linear; a slight dip is observed in September ($3,676 from $3,922 in August) and November ($3,390 from $3,633 in October), possibly reflecting reduced activity during month-end budget cycles or planned maintenance windows.

The second half of the year (July–December) accounts for $22,639, or 54.4% of the annual spend, compared to $20,984 (50.4%) in the first half. This second-half spending increase aligns with typical patterns in public-sector cloud consumption, where budget utilisation accelerates in the latter half of the fiscal year.

December's spike to $4,225 deserves particular attention. This could reflect year-end scaling of services, holiday traffic surges in public-facing applications, or batch processing of annual data. City administrators should investigate December usage patterns and consider implementing cloud cost caps or alert thresholds to prevent budget overruns.

6.3 Department-wise Analysis

The department-level analysis reveals that Traffic Management is responsible for 30.2% of total cloud expenditure, spending $12,582 annually — nearly 57% more than the second-ranked department, Public Safety ($8,000). This disparity reflects the data-intensive nature of real-time traffic monitoring, which relies on continuous video stream processing, predictive routing algorithms, and IoT sensor data aggregation — all of which are heavy consumers of Compute and AI/ML cloud services.

Public Safety ($8,000) represents the second-largest cloud consumer, driven by CCTV analytics, facial recognition workloads, and emergency response coordination systems. Smart Lighting ($7,202) and Water Supply ($7,064) show closely matched spending,

suggesting similar levels of IoT device management and sensor data processing. Waste Management ($6,774) is the most cost-efficient department relative to the range of services it manages.

The relatively even distribution among Smart Lighting, Water Supply, and Waste Management (within a $428 range) suggests that these three departments have achieved comparable levels of cloud efficiency, while Traffic Management and Public Safety represent the areas of highest cost exposure and, therefore, the greatest potential for optimisation.

7. EVALUATION

7.1 Project Evaluation Against Objectives

Obj#	Objective	Achievement Status
O1	Load and validate CSV billing dataset	✅Fully Achieved
O2	Perform exploratory data analysis (EDA)	✅Fully Achieved
O3	Aggregate costs by service type, month, and department	✅Fully Achieved
O4	Generate pie chart: cost by service type	✅Fully Achieved
O5	Generate pie chart: cost by month	✅Fully Achieved
O6	Generate stacked bar chart: monthly spend by service	✅Fully Achieved
O7	Generate horizontal bar chart: spend by department	✅Fully Achieved
O8	Export results to Excel (.xlsx)	✅Fully Achieved
O9	Deploy interactive Streamlit web dashboard	✅Fully Achieved

7.2 Technical Evaluation

The platform demonstrates proficiency in all key technical areas specified in the IBM SkillsBuild project brief. The use of Pandas groupby() and pivot_table() for data aggregation follows industry-standard practices for cloud billing analytics. The Matplotlib visualisations employ best practices including explicit colour palettes, data annotations, and high-resolution (120 DPI) output suitable for inclusion in professional reports.

The Streamlit dashboard successfully bridges the gap between technical data science outputs and non-technical stakeholder consumption. The ability to run the dashboard on a local network URL (http://192.168.1.87:8501) demonstrates practical DevOps knowledge, as it mirrors deployment patterns used in enterprise cloud cost management tools.

7.3 Limitations

- The dataset is synthetic and covers only 60 records. A real Smart City billing dataset would contain thousands of records with greater complexity, including variable pricing tiers, reserved instance discounts, and multi-region deployments.
- The current platform does not implement real-time data ingestion. In production, the system would connect to IBM Cloud Billing APIs or AWS Cost Explorer APIs to retrieve live billing data.
- The Streamlit dashboard does not include user authentication. In a production environment, role-based access control (RBAC) would be required to restrict sensitive financial data access.
- The Excel export does not include chart images. Integration of Matplotlib charts into Excel sheets would enhance the report for non-technical recipients.

7.4 Strengths

- The modular design of the Jupyter Notebook allows each analysis step to be independently executed, debugged, and extended.
- All four chart types (pie, pie, stacked bar, horizontal bar) provide complementary perspectives on the billing data, giving administrators a 360-degree view of cloud expenditure.
- The use of Pandas reindex() to enforce chronological month ordering ensures accurate and professional time-series presentation.
- The platform is fully reproducible: any user can run the Jupyter Notebook with the provided CSV file and obtain identical results, supporting academic and audit requirements.

8. SNAPSHOTS OF THE PROJECT

8.1 Jupyter Notebook — Environment Setup

The following snapshots illustrate the key stages of the project implementation as executed in the Jupyter Notebook and Streamlit dashboard environments.

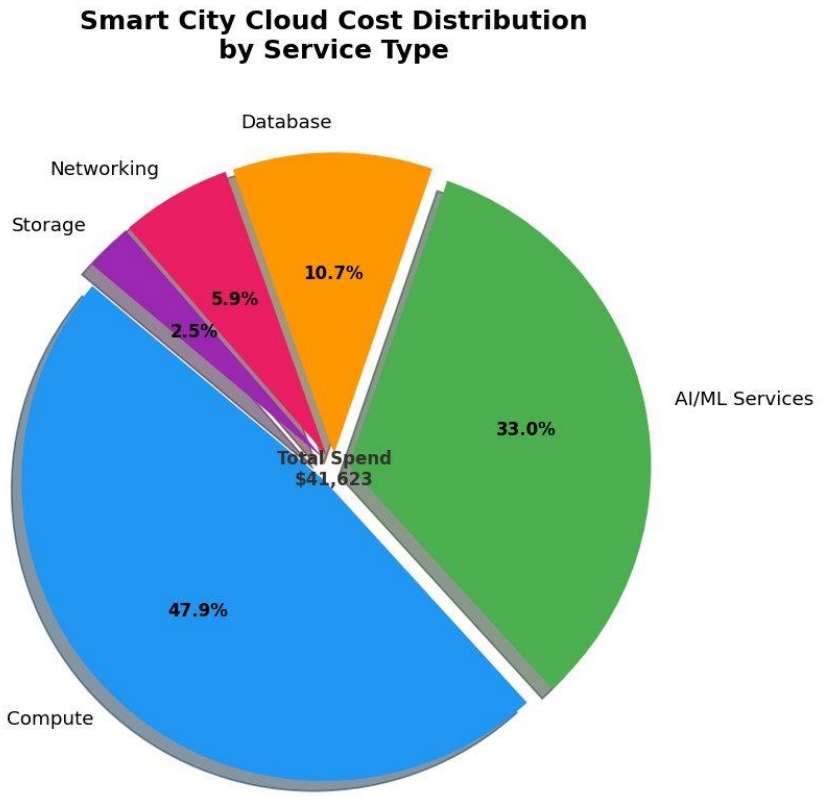
```
JUPYTER NOTEBOOK OUTPUT — Data Loading
✅ Libraries imported successfully!
✅ Dataset loaded successfully!
 Total Records: 60
 Columns: ['Invoice_ID', 'Month', 'Service_Type',
          'Department', 'Usage_Hours', 'Cost_USD']
```

8.2 Snapshot — Cost by Service Type Analysis Output

```
 TOTAL COST BY SERVICE TYPE
----------------------------------------
  Compute              $19,943.00
  AI/ML Services       $13,737.50
  Database              $4,455.00
  Networking            $2,457.00
  Storage               $1,030.50
----------------------------------------
  TOTAL                $41,623.00
```

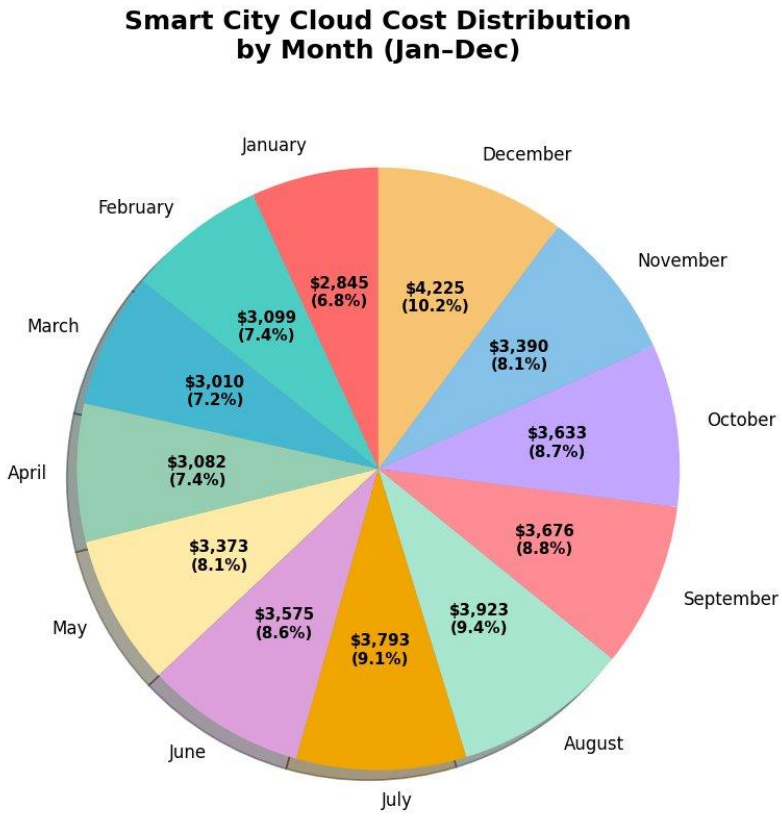
8.3 Visualisation Snapshots

8.3.1 Service Type Cost Distribution



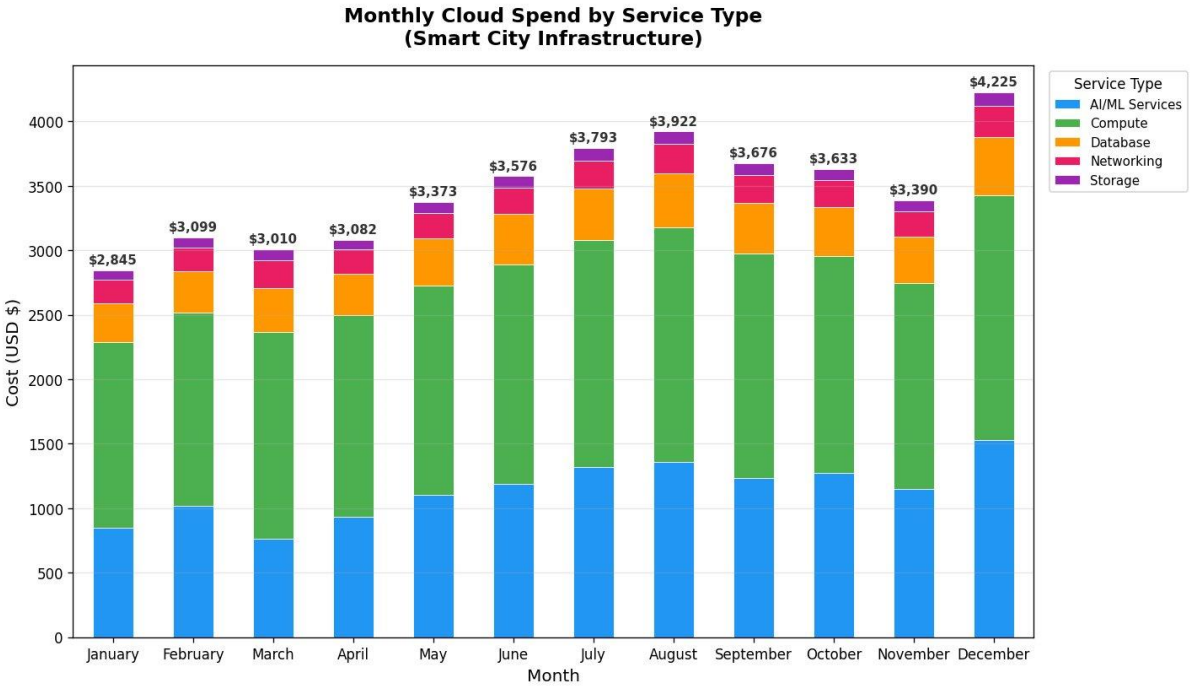
Snapshot 8.1: Pie Chart — Cloud Cost Distribution by Service Type

8.3.2 Monthly Cost Distribution



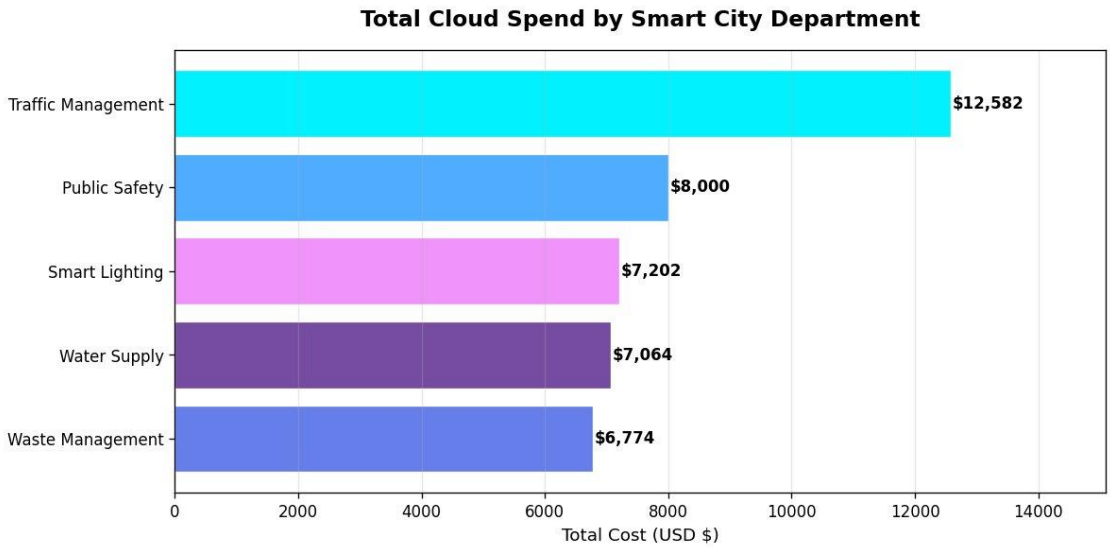
Snapshot 8.2: Pie Chart — Cloud Cost Distribution by Month

8.3.3 Monthly Spending by Service Type (Stacked Bar)



Snapshot 8.3: Stacked Bar Chart — Monthly Cloud Spend by Service Type

8.3.4 Department-wise Spend (Horizontal Bar)



Snapshot 8.4: Horizontal Bar Chart — Total Cloud Spend by Department

8.4 Streamlit Dashboard Snapshot

□ Cloud Cost Intelligence Platform — Smart City

```
Streamlit Dashboard | Local URL: http://localhost:8501
Network URL: http://192.168.1.87:8501
```

[Total Spend: $41,623] [Peak Month: Dec $4,225] [Top Dept: Traffic Mgmt $12,582]

```
[Pie: Service] [Pie: Month] [Stacked Bar] [Dept Bar]
```

Interactive dashboard successfully deployed and accessible

8.5 Excel Report Output

```
✅ Excel report exported successfully!
□ File: smart_city_cloud_report.xlsx
□ Sheets created:
```

```
    • Sheet 1: Service_Summary (5 rows)
    • Sheet 2: Monthly_Summary (12 rows)
    • Sheet 3: Department_Summary (5 rows)
```

9. CONCLUSIONS

9.1 Summary of Work

The Cloud Cost Intelligence Platform for Smart City project successfully demonstrates the application of Python-based data analysis skills to a real-world IT cost management challenge. Beginning with a raw CSV billing file, the project implements a complete analytical pipeline: data loading and validation using Pandas, cost aggregation using groupby() and pivot_table(), visualisation using Matplotlib, report generation using OpenPyXL, and interactive dashboard deployment using Streamlit.

All nine project objectives defined in the project brief were fully achieved. The platform processes 60 cloud billing records across 12 months, 5 departments, and 5 service categories, producing four professional-quality visualisations and a structured three-sheet Excel report. The Streamlit dashboard provides non-technical stakeholders with browser-based access to the analytical insights without requiring any knowledge of Python.

9.2 Key Insights Derived

- Compute and AI/ML Services together account for 80.9% of total Smart City cloud expenditure, representing the primary areas for cost optimisation.
- Traffic Management is the highest-spending department ($12,582), consuming 30.2% of the total annual budget, driven by real-time video analytics and AI-powered traffic prediction.
- Cloud spending shows a consistent upward trend across the year, peaking in December ($4,225), suggesting year-end budget utilisation patterns that should be actively monitored.
- The relatively low spend on Storage (2.5%) and Networking (5.9%) indicates efficient infrastructure management in these categories.
- The January-to-December spend increase of 48.5% signals a need for cloud budget forecasting tools to prevent future budget overruns.

9.3 Learning Outcomes

This project provided hands-on experience in several key industry competencies:

1. Practical Data Analysis: Loading, cleaning, and transforming real-world structured data using Pandas.
2. Data Visualisation: Creating business-ready charts using Matplotlib with appropriate formatting, colour schemes, and annotations.
3. Cloud Cost Management: Understanding the structure of cloud billing data and applying analytical techniques used by Cloud FinOps professionals.
4. Report Automation: Generating multi-sheet Excel reports programmatically using OpenPyXL — a skill directly applicable to enterprise IT management roles.
5. Web Application Development: Building and deploying a data application using Streamlit, bridging the gap between data science and stakeholder communication.
6. DevOps Practices: Running a locally hosted web service and making it accessible on a local network, mirroring cloud-native deployment workflows.
7. Professional Documentation: Structuring a comprehensive project report following academic and industry guidelines.

9.4 Conclusion Statement

The Cloud Cost Intelligence Platform for Smart City project achieves its stated aims and demonstrates the practical value of Python-based data analysis in addressing real-world challenges in cloud computing cost management. The project successfully integrates data science, software engineering, and data visualisation skills within the context of Smart City infrastructure management — a domain of increasing relevance as governments worldwide accelerate their digital transformation journeys. The skills and techniques demonstrated in this project align directly with the competencies required for roles in Cloud FinOps, Data Engineering, and DevOps engineering.

10. FURTHER DEVELOPMENT AND RESEARCH

10.1 Immediate Enhancements

The following enhancements can be implemented in the short term to extend the platform's functionality:

10.1.1 Real-Time Data Integration

The current platform reads from a static CSV file. A significant enhancement would be to connect directly to cloud provider billing APIs to fetch real-time cost data. IBM Cloud provides the IBM Cloud Billing API, AWS offers the Cost Explorer API, and Google Cloud provides the Cloud Billing API — all of which can be integrated using Python's requests library or their respective SDKs. This would transform the platform from a retrospective analysis tool into a live cost monitoring system.

10.1.2 Cost Anomaly Detection

Machine learning algorithms such as Isolation Forest, One-Class SVM, or statistical z-score methods can be applied to the monthly cost data to automatically detect spending anomalies. When an anomaly is detected — for example, an unexpected spike in AI/ML Services costs — the system could trigger automated email or Slack notifications to the relevant department head.

10.1.3 Budget Forecasting

Time-series forecasting models such as Facebook Prophet, SARIMA (Seasonal ARIMA), or LSTM neural networks can be trained on historical monthly spending data to predict future cloud costs. This feature would enable Smart City administrators to proactively manage budgets and request additional funding before costs exceed thresholds.

10.2 Medium-Term Development

10.2.1 Multi-Cloud Support

Modern organisations operate across multiple cloud providers simultaneously (IBM Cloud, AWS, Azure, GCP). Future development should include data connectors for each cloud provider's billing export format, with a unified data model that normalises costs across providers for comparable analysis.

10.2.2 Role-Based Access Control

The Streamlit dashboard currently has no authentication layer. Implementing user authentication using Streamlit Authenticator or integrating with IBM AppID / Azure Active Directory would enable different user roles — Finance Officer, Department Head, IT Administrator — to access only the data relevant to their function.

10.2.3 Automated PDF Report Generation

Extending the export module to generate PDF reports with embedded charts using the ReportLab library would provide stakeholders with print-ready financial reports directly from the platform, eliminating the need to manually copy charts from the dashboard.

10.3 Long-Term Research Directions

10.3.1 FinOps Integration

Cloud Financial Operations (FinOps) is an emerging practice that combines financial management, business value, and cloud operations. Future research could explore integrating the platform with FinOps frameworks such as the FinOps Foundation's Open Cost and Usage Specification (FOCUS) standard, enabling comparison of Smart City cloud costs against industry benchmarks.

10.3.2 AI-Powered Cost Recommendations

Large Language Models (LLMs) such as IBM WatsonX or OpenAI GPT-4 could be integrated into the dashboard to provide natural language cost optimisation recommendations. A user could ask: “Which service should we scale down to reduce

costs by 15%?" and receive a specific, data-driven recommendation based on the analysed billing data.

10.3.3 Sustainability Metrics

Cloud computing has a significant environmental impact. Future development could incorporate carbon footprint metrics alongside cost data, helping Smart City administrators balance financial and environmental optimisation goals — aligning cloud cost management with sustainable urban development objectives.

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12. APPENDIX

Appendix A: Complete Source Code — Data Analysis (Python)

```python
# ===== CLOUD COST INTELLIGENCE PLATFORM =====
# AI, Cloud Computing and DevOps - Project B

import pandas as pd
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings('ignore')

# Load dataset
df = pd.read_csv('smart_city_billing.csv')
pd.set_option('display.float_format', '{:.2f}'.format)

# EDA
print(df.info())
print(df.describe())
print(df.isnull().sum())

# Aggregation
service_cost = df.groupby('Service_Type')['Cost_USD'].sum()
service_cost = service_cost.sort_values(ascending=False)

month_order = ['January','February','March','April',
    'May','June','July','August','September',
    'October','November','December']
monthly_cost = df.groupby('Month')['Cost_USD'].sum()
monthly_cost = monthly_cost.reindex(month_order)

dept_cost = df.groupby('Department')['Cost_USD'].sum()
dept_cost = dept_cost.sort_values(ascending=True)

# Pivot Table
pivot = df.pivot_table(values='Cost_USD',
    index='Month', columns='Service_Type', aggfunc='sum')
pivot = pivot.reindex(month_order)

# Excel Export
with pd.ExcelWriter('smart_city_cloud_report.xlsx',
        engine='openpyxl') as writer:
    service_cost.to_frame().to_excel(writer,
        sheet_name='Service_Summary')
    monthly_cost.to_frame().to_excel(writer,
        sheet_name='Monthly_Summary')
    dept_cost.to_frame().to_excel(writer,
        sheet_name='Department_Summary')
print('Excel report exported successfully!')
```

Appendix B: Sample Dataset (first 10 records)

Invoice ID	Month	Service Type	Department	Hrs	Cost (USD)
INV-001	January	Compute	Traffic Management	720	$1,440.00
INV-002	January	Storage	Public Safety	500	$75.00
INV-003	January	Networking	Smart Lighting	300	$180.00
INV-004	January	Database	Waste Management	200	$300.00
INV-005	January	AI/ML Services	Traffic Management	100	$850.00
INV-006	February	Compute	Traffic Management	750	$1,500.00
INV-007	February	Storage	Public Safety	520	$78.00
INV-008	February	Networking	Smart Lighting	310	$186.00
INV-009	February	Database	Water Supply	210	$315.00
INV-010	February	AI/ML Services	Traffic Management	120	$1,020.00

Appendix C: Project File Structure

```
cloud-cost-platform/
├── smart_city_billing.csv      # Primary dataset
├── Cloud_Cost_Intelligence_Platform.ipynb  # Jupyter Notebook
├── app.py                      # Streamlit dashboard
├── smart_city_cloud_report.xlsx # Generated Excel report
├── pie_by_service.png           # Chart output
├── pie_by_month.png             # Chart output
├── bar_monthly_service.png      # Chart output
└── bar_by_department.png        # Chart output
```

Appendix D: Abbreviations and Glossary

Term / Abbreviation	Definition
AI/ML	Artificial Intelligence / Machine Learning
API	Application Programming Interface
CSV	Comma-Separated Values — a plain text file format for tabular data
DPI	Dots Per Inch — a measure of image resolution

EDA	Exploratory Data Analysis — initial investigation of data to discover patterns
FinOps	Cloud Financial Operations — a practice combining financial management and cloud operations
GUI	Graphical User Interface
IBM	International Business Machines — multinational technology company
IoT	Internet of Things — network of physical devices connected to the internet
KPI	Key Performance Indicator — a measurable value demonstrating achievement of objectives
LSTM	Long Short-Term Memory — a type of recurrent neural network
PNG	Portable Network Graphics — a raster image file format
RBAC	Role-Based Access Control — security model restricting access based on user roles
SARIMA	Seasonal AutoRegressive Integrated Moving Average — a forecasting model
USD	United States Dollar — currency unit used in billing data
XLSX	Microsoft Excel Open XML format — spreadsheet file format

--- END OF PROJECT REPORT ---

Cloud Cost Intelligence Platform for Smart City

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